FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Growth & Development

Lecture | Week 6 · Tetanus, skin, poisons & hormones | Paediatrics | Dr Nabbale

Growth (weight, length/height, head circumference, MUAC, plotted as WHO Z-scores) and development (gross motor, fine motor and vision, speech, language and hearing, and social–emotional–self-care skills) are the vital signs of childhood; Edith must know the normal patterns, the limit ages and red flags, how to plot and interpret a growth chart, Tanner staging, and how to assess and act on a child who is faltering or delayed.

Must know before the ward round

1. Weight: birth ~3.2 kg, may lose up to 10% in the first week, back to birthweight by 2–3 weeks (NICE: by 3 weeks); doubles by ~5 months, triples by 12 months, quadruples by ~2.5 years.
2. Length: 50 cm at birth, +25 cm in year 1 (75 cm), +12 cm in year 2 (~87 cm), doubles to 100 cm by ~4 years, then ~5–6 cm/year until puberty.
3. Head circumference: 35 cm at birth, ~46 cm at 1 year (about +12 cm: 2 cm/month for 3 months, 1 cm/month to 6 months, 0.5 cm/month to 12 months). Microcephaly < −2 SD; macrocephaly > +2 SD; crossing centiles matters more than one value.
4. WHO Z-scores: < −2 = moderate, < −3 = severe. WAZ = underweight, HAZ/LAZ = stunting (chronic), WHZ/WLZ = wasting (acute); WHZ > +2 overweight, > +3 obese (under 5).
5. SAM (WHO 2023, UCG 2023): WHZ < −3 or MUAC < 11.5 cm (6–59 months) or bilateral pitting oedema. MAM: WHZ −3 to −2 or MUAC 11.5–12.4 cm. MUAC only from 6 months.
6. Limit ages (refer if not achieved): social smile 8 weeks; fixes and follows 3 months; head control 4 months; reaches for objects 6 months; sits unsupported 9 months; pincer grip 12 months; walks alone 18 months; 6+ words 18 months; 2-word phrases 2 years (by 2.5 years at the latest).
7. Red flags at any age: LOSS of skills, parental concern, no response to sound or failure to fix and follow, asymmetry or hand preference before 12 months, persistent primitive reflexes after 6 months, abnormal tone, abnormal head growth.
8. Correct for prematurity until 2 years: corrected age = chronological age − weeks born early.
9. Primitive reflexes: Moro, rooting, palmar grasp, ATNR, stepping disappear by 3–6 months; parachute appears 6–9 months and stays.
10. Puberty: girls thelarche at ~10–11 y (8–13), menarche ~2–2.5 years later at B4; boys testicular volume ≥4 mL at ~11–12 y (9–14), PHV at G4. Precocious: girls < 8 y, boys < 9 y. Delayed: no breast buds by 13 y (girls) or testes < 4 mL by 14 y (boys).
11. Mid-parental height: boys (father + mother + 13)/2; girls (father + mother − 13)/2; target range ± 8.5 cm.
12. Uganda: weigh, measure length/height and MUAC at every contact and plot on the Child Health Card; test every child with growth faltering or developmental delay for HIV (UCG 2023); vitamin A (100,000 IU at 6–11 months, 200,000 IU at 12–59 months) and albendazole (200 mg 1–2 y, 400 mg >2 y) every 6 months.

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In a nutshell

Growth is the increase in size (weight, length/height, head circumference); **development** is the acquisition of skills (moving, using hands and eyes, communicating, relating to people and looking after oneself). Both follow predictable patterns, and both are **sensitive markers of a child's health**: a child who is ill, hungry, unloved, infected with HIV or TB, or has a brain or genetic problem will often show it first as a **flattening growth curve** or **delayed milestones**.

In Uganda, about **1 in 4 children under five is stunted**, and developmental delay is common because of birth asphyxia, prematurity, malnutrition, malaria, meningitis, HIV and limited stimulation. Measuring and plotting every child, asking every mother about milestones, and acting early are among the most powerful things a clinician can do.

> **CLINICAL PEARL**
> The one thing never to forget: **a single measurement means little; the trend is everything.** Weigh and measure accurately, plot on the right WHO chart (sex, age, measurement), and ask "is this child following their curve?" Likewise for development, the most worrying sign is not slowness but **regression**: a child who **loses** a skill needs urgent assessment.

Definitions & classification

Term	Definition
Growth	Increase in physical size; measured as weight, length (lying, <2 years or <87 cm) or height (standing, ≥2 years), head circumference (HC), MUAC
Development	Acquisition of function across domains: **gross motor**, **fine motor and vision**, **speech, language and hearing**, **social, emotional and self-care** (and cognition)
Z-score (SD score)	How many standard deviations a measurement lies from the WHO median for age and sex. Z = (observed − median) ÷ SD. 0 = median; −2 ≈ 2nd–3rd centile; +2 ≈ 97th–98th centile
Centile	The percentage of the reference population below the measurement (e.g. 50th = median)
Underweight	Weight-for-age (WAZ) < −2 (severe < −3)
Stunting	Length/height-for-age (LAZ/HAZ) < −2 (severe < −3): **chronic** undernutrition and illness
Wasting	Weight-for-length/height (WLZ/WHZ) < −2 (severe < −3) or MUAC < 12.5 cm: **acute** undernutrition
SAM (WHO 2023 guideline on wasting and nutritional oedema; UCG 2023)	WHZ < −3, **or** MUAC < 11.5 cm (6–59 months), **or** bilateral pitting oedema
MAM	WHZ −3 to < −2, or MUAC 11.5–12.4 cm
Overweight / obesity (under 5)	WHZ > +2 / > +3 (BMI-for-age above 5 years: > +1 overweight, > +2 obese)

Term	Definition
Microcephaly / macrocephaly	HC < −2 SD (severe < −3 SD) / HC > +2 SD
Faltering growth (NICE NG75)	Fall across ≥1 weight centile space if birthweight < 9th centile; ≥2 spaces if 9th–91st; ≥3 spaces if > 91st; or weight < 2nd centile at any time
Developmental delay	Milestones achieved significantly later than expected (beyond the limit age) in one domain
Global developmental delay	Significant delay in **two or more domains** in a child under 5 years
Developmental quotient (DQ)	Developmental age ÷ chronological (corrected) age × 100. DQ < 70 = significant delay
Regression	Loss of previously acquired skills: always abnormal
Corrected age	For preterm infants (until 2 years): chronological age minus the number of weeks born before 40 weeks

Epidemiology & Uganda context

- **Undernutrition**: the Uganda Demographic and Health Survey 2022 reported roughly **a quarter of under-fives stunted**, about **3–4% wasted** and about **1 in 10 underweight**, with higher rates in Karamoja, the West Nile and South-Western regions (stunting in the South-West despite food production), and among children of poorer, less-educated mothers.
- **Low birthweight** (<2.5 kg) affects about 1 in 10 births; prematurity and fetal growth restriction start many children on a lower curve.
- **Developmental delay**: global estimates suggest about **250 million children under 5** in low- and middle-income countries are at risk of not reaching their developmental potential because of poverty, stunting and lack of stimulation (Lancet series). In Uganda, birth asphyxia, kernicterus, cerebral malaria, meningitis, HIV and severe malnutrition add neurological injury.
- **Services**: growth monitoring is meant to happen at every immunisation and sick-child visit using the **Child Health Card** (weight-for-age plotted; length and MUAC increasingly recorded). In practice, **length is often not measured**, MUAC tapes run out, and milestones are rarely asked about. **Early Childhood Development (ECD)** policy and **Nurturing Care** (WHO/UNICEF 2018: good health, adequate nutrition, responsive caregiving, safety and security, opportunities for early learning) frame national programmes.

UGANDA CONTEXT

UCG 2023 (HIV chapter) says: monitor growth with **weight, length/height and MUAC at every encounter** and record it on the growth card; MUAC only from **6 months**; **failure to gain weight or height may be the first sign of HIV** (failure to thrive affects up to 50% of HIV-infected children); assess **age-specific milestones** at each visit; **test children with developmental delay for HIV** and start ART; measure head circumference. Apply this to every child, not only HIV-exposed children.

Physiology & principles

Phases of growth (the infancy–childhood–puberty model)

1. **Fetal and infancy phase (0–2 years)**: fastest growth; driven mainly by **nutrition** (and insulin, IGF-2). Infants often **cross centiles** in the first 6–18 months to reach their genetic channel ("catch-up" or "catch-down"). Stunting that develops here (the **first 1000 days**, conception to 2 years) is largely irreversible.
2. **Childhood phase (2 years–puberty)**: steady **5–6 cm/year**; driven by **growth hormone** and **thyroid hormone**. A child should follow their centile; crossing down suggests disease.

3. **Pubertal phase**: **sex steroids** plus GH → the **pubertal growth spurt** (peak height velocity about 8–9 cm/year in girls, 9–10 cm/year in boys), then fusion of the epiphyses.

Why measurements differ in what they tell you

- **Weight** reflects recent nutrition and illness (and fluid: oedema can hide wasting).
- **Length/height** reflects long-term nutrition, chronic disease, endocrine and genetic factors. It falls more slowly than weight.
- **Weight-for-height** separates the thin child (wasted) from the short child (stunted but proportionate).
- **Head circumference** reflects **brain growth** in the first 2 years. A falling HC centile is a sign of brain injury or disease (HIV encephalopathy, Rett syndrome, severe malnutrition); a rising one suggests hydrocephalus or a mass.
- **MUAC** reflects muscle and fat mass; a MUAC below 11.5 cm independently predicts death in children 6–59 months, which is why it is used for community screening.

Principles of development

- **Cephalocaudal and proximodistal**: control develops from **head to toe** (head control → sitting → standing) and from **trunk to fingertips** (whole-hand palmar grasp → pincer grip).
- **Primitive reflexes must disappear** (cortical inhibition) before voluntary skills emerge; **protective reflexes** (parachute) appear.
- **Sequence is fixed, rate varies**: some children bottom-shuffle and walk late (often familial) but the sequence is the same.
- **Domains interact**: a deaf child's language is delayed; a blind child's motor skills may be delayed; a child with CP may have normal cognition hidden behind motor and speech problems.
- **Environment matters**: responsive caregiving, talking, play and good nutrition improve outcomes, especially in the first 3 years when brain plasticity is greatest.

Normal growth: numbers to know

Weight, length and head circumference

Age	Weight (approx.)	Length/height	Head circumference
Birth	3.2–3.5 kg (normal 2.5–4.0)	50 cm	35 cm
First week	Loses up to 7–10%		
2–3 weeks	Back to birthweight		
0–3 months	Gains ~25–30 g/day (~200 g/week)	+3–4 cm/month	+2 cm/month
4–5 months	**Doubles** birthweight (~6.5 kg)		+1 cm/month (4–6 months)
12 months	**Triples** (~9.5–10 kg)	**75 cm** (+25 cm)	**~46 cm** (+0.5 cm/month 7–12 months)
2 years	~12 kg	**~87 cm** (+12 cm); ≈ half adult height	~48 cm
2.5 years	**Quadruples** (~13 kg)		
4 years	~16 kg	**~100 cm** (double birth length)	~50 cm
4 years to puberty	+2–3 kg/year	**+5–6 cm/year**	Slow

Estimating weight in an emergency (APLS; use a real weight whenever possible): 0–12 months: (0.5 × age in months) + 4; 1–5 years: (2 × age in years) + 8; 6–12 years: (3 × age in years) + 7.

Other physical milestones

Feature	Normal
Posterior fontanelle	Closes by **6–8 weeks**
Anterior fontanelle	Closes at **12–18 months** (up to 24). Large or late: rickets, hypothyroidism, hydrocephalus, Down syndrome, osteogenesis imperfecta. Early: craniosynostosis, microcephaly. **Bulging**: raised ICP (meningitis); **sunken**: dehydration
Teeth	First tooth (lower central incisor) at about **6 months** (4–10); 20 primary teeth by **2.5–3 years**; first permanent molars ~6 years. No teeth by 13–15 months: think hypothyroidism, rickets, hypopituitarism, genetic syndromes (or familial)
Vital signs (APLS ranges, awake)	HR: <1 y 110–160; 1–2 y 100–150; 2–5 y 95–140; 5–12 y 80–120. RR: <1 y 30–40; 1–2 y 25–35; 2–5 y 25–30; 5–12 y 20–25. SBP: <1 y 70–90; 1–2 y 80–95; 2–5 y 80–100; 5–12 y 90–110 mmHg
Fast breathing (WHO/IMCI)	≥60/min under 2 months; ≥50/min at 2–12 months; ≥40/min at 1–5 years

Normal development: milestones by domain

Median ages are approximate; the **limit age** is the age by which about 95–97% of children have achieved the skill (refer if not achieved).

Gross motor

Skill	Median age	Limit age
Lifts head when prone (45°)	6 weeks–3 months	
Good head control (no head lag on pull-to-sit)	3–4 months	**4 months**
Rolls front to back / back to front	4–6 months	
Sits without support (back straight)	6–8 months	**9 months** (NICE CP warning: 8 months corrected)
Crawls (or bottom-shuffles)	8–9 months	
Pulls to stand, cruises around furniture	9–11 months	
Stands alone	12 months	
Walks alone	12–13 months	**18 months**
Runs; climbs onto furniture	18 months–2 years	
Kicks a ball; jumps with both feet	2 years	
Stairs: alternating feet going up; rides a tricycle	3 years	
Hops on one foot; stairs adult fashion	4 years	
Skips	5 years	

WHO Multicentre Growth Reference Study windows (healthy breastfed children, 1st–99th centile): sitting without support 3.8–9.2 months; standing alone 6.9–16.9 months; **walking alone 8.2–17.6 months**.

Fine motor and vision

Skill	Median age	Limit age
Fixes and follows a face or object	6 weeks	**3 months**
Hands come together in the midline; holds a rattle	3–4 months	
Reaches for objects (palmar grasp)	4–5 months	**6 months**
Transfers objects hand to hand	6–7 months	**9 months**
Inferior pincer, then **neat pincer grip** (bean, raisin)	9–12 months	**12 months**
Bangs two cubes together; points with index finger; casts objects	10–12 months	
Builds a tower of 3 cubes; scribbles; turns several pages	18 months	
Tower of 6–7 cubes; draws a vertical line; turns single pages	2 years	
Tower of 9; copies a **circle**; builds a bridge	3 years	
Copies a **cross** (+) (4 y), a **square** (4.5 y)	4 years	
Copies a **triangle**; draws a person with several parts	5 years	

> NOTE
> **Hand preference before 12 months is abnormal**: it suggests weakness of the other hand (hemiplegic CP, brachial plexus injury). Handedness is usually established by 2–3 years.

Speech, language and hearing

Skill	Median age	Limit age
Startles to loud sound	Newborn	
Coos; turns towards sound/voice	3–4 months	
Polysyllabic babble ("ba-ba-da")	6–7 months	7 months
Consonant babble ("mama", "dada" non-specifically); responds to name	8–10 months	10 months
Understands "no"; 1–3 words with meaning; waves bye-bye	12 months	
6–10 words with meaning; points to body parts; follows one-step commands	18 months	**18 months (6 words)**
Two-word phrases ("mama come"); 50+ words	2 years	**2 years** (and 2.5 years at the latest)
3–4-word sentences; knows name, age, sex; asks "what/who" questions	3 years	2.5–3 years
Tells stories; asks "why"; understandable to strangers; counts	4 years	
Fluent speech; knows colours; can define words	5 years	

Social, emotional and self-care

Skill	Median age	Limit age
Social smile	6 weeks	**8 weeks**
Laughs; enjoys social play	3–4 months	
Puts food/objects in mouth; stranger awareness begins	6–7 months	

Skill	Median age	Limit age
Stranger anxiety; plays peek-a-boo	9–10 months	10 months
Waves bye-bye; plays pat-a-cake; drinks from a cup with help	12 months	
Feeds self with spoon (messy); **points to show interest** (joint attention); copies chores	18 months	18 months
Symbolic (pretend) play; parallel play; dry by day (variable)	2 years	2–2.5 years
Interactive play, takes turns, shares; dresses with help; dry at night (variable)	3 years	3–3.5 years
Has friends; dresses and undresses; toilets independently	4 years	
Plays by rules; understands fairness	5 years	

EXAM FAVOURITE

The **CDC (2022 revision)** "Learn the Signs. Act Early." checklists list skills that **most (about 75%) children** have by each age, e.g. at 9 months: sits without support, uses fingers to rake food, looks for objects dropped out of sight, babbles "mamamama"; at 12 months: pulls to stand, walks holding furniture, pincer grip, waves bye-bye, says "mama/dada" with meaning; at 18 months: walks without holding on, scribbles, tries to say 3 or more words, follows one-step directions without gestures, points to show something interesting. Not reaching a CDC milestone is a reason to **look more closely**; not reaching a **limit age** is a reason to **refer**.

Primitive and protective reflexes

Reflex	How to elicit	Appears	Disappears
Moro	Sudden head drop (supported) → arms abduct and extend, then adduct	Birth (from 28–32 weeks)	**3–6 months** (asymmetry: brachial plexus palsy, fractured clavicle, hemiplegia)
Rooting / sucking	Stroke cheek → turns to that side and sucks	Birth	3–4 months
Palmar grasp	Finger in palm → grasps	Birth	**3–4 months** (must go before voluntary reach)
Plantar grasp	Pressure on sole → toes curl	Birth	9–12 months
Asymmetric tonic neck reflex (ATNR)	Turn head → "fencing posture"	2 weeks–1 month	**4–6 months**; obligatory or persistent = CP
Stepping / placing	Hold upright, feet touch surface	Birth	2 months
Parachute (protective)	Tip child forward towards a surface → arms extend	**6–9 months**	**Persists for life** (absent/asymmetric = CP, hemiplegia)

Puberty

Tanner (sexual maturity) stages

Stage	Girls: breasts (B)	Boys: genitalia (G)	Both: pubic hair (PH)
1	Prepubertal (papilla only)	Prepubertal; testes **< 4 mL**	None
2	**Breast bud**: small mound, areola widens	Testes **≥ 4 mL**, scrotum enlarges and reddens	Sparse, long, slightly pigmented, at base of penis/labia

Stage	Girls: breasts (B)	Boys: genitalia (G)	Both: pubic hair (PH)
3	Breast and areola enlarge further, no separation of contours	Penis lengthens; testes 6–12 mL	Darker, coarser, curly, spreads over pubis
4	**Areola and papilla form a secondary mound**	Penis broadens, glans develops; testes 12–15 mL; scrotum darkens	Adult type, not on thighs
5	Adult contour; areola recedes to breast contour	Adult; testes 15–25 mL	Adult, spreads to medial thighs

Testicular volume is measured with a **Prader orchidometer** (beads 1–25 mL).

Sequence and timing

	Girls	Boys
First sign	**Thelarche** (breast bud, B2): mean ~10–11 years (range 8–13)	**Testicular enlargement ≥ 4 mL** (G2): mean ~11–12 years (range 9–14)
Then	Pubic hair (pubarche), growth spurt	Pubic hair, penile growth
Peak height velocity	**Early** (B2–B3), ~11.5–12 years, ~8–9 cm/year	**Late** (G4), ~13.5–14 years, ~9–10 cm/year (why men are taller)
Late	**Menarche** at B4, **~2–2.5 years after thelarche** (mean ~12.5–13 y); height gain after menarche only ~5–7 cm	Voice breaks, facial and axillary hair, adult body habitus
Precocious	Signs before **8 years** (some US guidance uses 7 in White and 6 in Black girls)	Before **9 years**
Delayed	No breast development by **13 years**, or no menarche by **15 years** (or 3 years after thelarche)	Testicular volume < 4 mL at **14 years**

> **NOTE**
> Precocious puberty in **boys** is more often pathological (CNS tumour, hamartoma, hydrocephalus, congenital adrenal hyperplasia) than in girls (mostly idiopathic central). **Delayed puberty** is most often **constitutional delay of growth and puberty** (often familial, delayed bone age), but in Uganda think of **chronic disease and undernutrition** (HIV, sickle cell disease, TB, IBD, chronic kidney disease), hypothyroidism, and gonadal failure (Turner syndrome in a short girl, Klinefelter in a tall boy).

Clinical acumen

Focused history

For growth

- **Birth**: gestation, birthweight (LBW, SGA), multiple birth.
- **Feeding history** (the single most useful part): **exclusive breastfeeding to 6 months?** How often? Formula or cow's milk (how prepared, dilution)? **Complementary foods** from 6 months: what, how thick, how often, who feeds, animal-source foods? Appetite, feeding difficulties, food insecurity. Use a **24-hour dietary recall**.
- **Illness**: diarrhoea, recurrent infections, cough >2 weeks (TB), fevers (malaria), vomiting, polyuria, snoring, previous admissions.
- **HIV**: mother's status, PMTCT, child's test results (DNA PCR, antibody), ART and co-trimoxazole.
- **Family**: parents' heights (mid-parental height), siblings' growth, deaths of siblings, consanguinity, family history of delayed puberty.

- **Social**: who cares for the child, mother's health and mood, alcohol, poverty, recent separation, child abuse/neglect risk.
- **Immunisation** (UNEPI card), **vitamin A** and **deworming** in the last 6 months.

For development

- **Parental concern**: "What worries you?" Parents are usually right.
- Milestones **by domain**: when did the child smile, hold the head, sit, crawl, walk, say first words, put words together? What can the child do **now**? Use the Child Health Card if recorded.
- **Any loss of skills?**
- **Hearing and vision**: does the child respond to voices, look at faces, recognise people across the room?
- **Antenatal** (infections, alcohol, drugs, HIV/syphilis), **perinatal** (prematurity, asphyxia, jaundice, neonatal admission, seizures) and **postnatal** (meningitis, cerebral malaria, head injury, seizures) history.
- **Play and stimulation**: who plays and talks with the child, toys, books, screen time, day-care.
- **Behaviour**: eye contact, pointing, response to name, repetitive behaviours (autism); sleep; feeding.
- **Family history** of delay, learning difficulty, consanguinity.

Examination

1. **Measure accurately** (see Clinical workup), plot, calculate Z-scores, compare with previous points.
2. **General**: wasting (buttocks, ribs), oedema (feet first: SAM), hair and skin changes, pallor, dysmorphic features (Down syndrome, fetal alcohol), signs of HIV (thrush, lymphadenopathy, parotid swelling), rickets (wrist widening, rosary, bowing), thyroid, bruises (abuse).
3. **Systemic**: chest (TB, recurrent pneumonia), heart murmurs (CHD), abdomen (organomegaly), genitalia and **Tanner staging** in older children.
4. **Developmental assessment**: observe play first, then test each domain with simple equipment (see Procedure below).
5. **Neurology**: tone, power, reflexes, primitive reflexes, head circumference, fontanelle, gait.
6. **Hearing and vision** screen.

Red flags

RED FLAG

Developmental red flags (refer for full assessment):

- **Loss of skills** at any age
- **Parental concern** about vision, hearing or development
- No **social smile by 8 weeks**; not **fixing and following by 3 months**; no response to sound
- **Asymmetry** of movement or **hand preference before 12 months**
- **Persistent primitive reflexes** beyond 6 months; very floppy or stiff
- Not **sitting by 9 months**, not **walking by 18 months**, no **words by 18 months**, no 2-word phrases by 2–2.5 years
- No pointing, poor eye contact, no response to name by 12–18 months (autism)
- **Head circumference** above +2 or below −2 SD, or crossing centiles

Growth red flags: weight < −3 Z or MUAC < 11.5 cm or oedema (SAM: treat now); falling across centile spaces; **height crossing downwards** after age 2; short stature with disproportion, dysmorphism or delayed puberty; HC falling.

Thinking like a clinician

- **Short but growing normally parallel to the 2nd centile, with short parents** = familial short stature (normal velocity, bone age = chronological age). **Short with delayed bone age and a parent who was a "late bloomer"** = constitutional delay. **Short and crossing centiles down** = pathology (hypothyroidism, GH deficiency, chronic disease, malnutrition, Turner syndrome).
- **Weight falls before height, then height follows** = undernutrition/illness. **Height falls first while weight is preserved (child looks chubby)** = endocrine (hypothyroidism, GH deficiency, Cushing's, including steroid use).
- **Oedema hides wasting**: a child with kwashiorkor may have a "normal" weight. Press the feet.
- **Plot against corrected age** for preterm babies until 2 years, or you will diagnose delay and faltering that are not there.
- **Isolated speech delay**: always check **hearing** first (chronic otitis media with effusion, meningitis sequelae), then consider autism, global delay, and lack of stimulation.
- **Motor delay with normal social and language skills** suggests CP, a neuromuscular disorder or benign late walking (bottom-shufflers). **Delay in all domains** suggests a global brain problem (genetic, HIE, infection, metabolic).
- **HIV**: growth faltering, developmental delay or regression, recurrent infections in a child of unknown status → **test**.
- **Measurement error** is the commonest cause of "abnormal" growth: check the scale (zeroed?), clothes removed, length measured lying with two people, correct chart for sex.

Clinical workup

Step 0: First 5 minutes

Usually a clinic task, but on the ward:

1. **ETAT**: emergency signs (airway, breathing, circulation, coma, convulsions, severe dehydration), then **priority signs** including **visible severe wasting** and **oedema of both feet**.
2. **SAM screen**: MUAC (6–59 months), oedema, WHZ. SAM with medical complications (danger signs, poor appetite, oedema +++, infants <6 months at risk) → **inpatient therapeutic care (ITC)**; check **glucose** (treat if < 3.0 mmol/L), **temperature** (hypothermia), dehydration (use ReSoMal per SAM protocol, not standard IV boluses).
3. **Measure properly** before anyone writes "normal growth".

Step 1: Focused history

- [ ] **Birthweight and gestation** → expected curve; corrected age.
- [ ] **Feeding** (breastfeeding, complementary diet, 24-hour recall) → commonest cause of faltering.
- [ ] **Illnesses** (diarrhoea, cough, fevers) → infection-driven faltering, TB, HIV.
- [ ] **HIV status** of mother and child → test if unknown.
- [ ] **Milestones by domain and any regression** → delay pattern.
- [ ] **Hearing and vision** → isolated delays.
- [ ] **Parents' heights, puberty timing** → genetic target.
- [ ] **Immunisation, vitamin A, deworming** → catch up.
- [ ] **Social situation** → neglect, food insecurity, maternal depression.

Step 2: Focused examination (with measurement technique)

Measurement	Technique
Weight	Calibrated scale zeroed; infant **naked** (or minimal clothing), older child light clothes, no shoes. Salter hanging scale or electronic scale; record to 0.1 kg (0.01 kg in infants on a baby scale)
Length (< 2 years or < 87 cm)	**Lying** on a length board, **two people**: one holds the head against the headboard (Frankfort plane vertical), the other straightens the knees and brings the footboard to the heels. Record to 0.1 cm
Height (≥ 2 years)	**Standing**, no shoes, heels, buttocks, shoulders and head touching the board, looking straight ahead. If a child ≥ 2 years is measured lying, subtract 0.7 cm (WHO)
Head circumference	Non-stretch tape around the **largest occipitofrontal circumference** (just above the eyebrows, around the occiput); take the largest of 3
MUAC (6–59 months)	**Left** arm, midpoint between acromion and olecranon (arm bent to find it), arm then hanging relaxed; tape snug but not compressing. Red < 11.5 cm, yellow 11.5–12.4, green ≥ 12.5
Oedema	Thumb pressure on the dorsum of **both feet** for 3 seconds; pit remains = bilateral pitting oedema (+ feet, ++ feet and legs/hands/arms, +++ generalised including face)

Then: systemic examination, Tanner stage (if > 8 years), developmental assessment, neurology, hearing and vision.

Step 3: Bedside tests

Test	What it tells you
Plot on the WHO chart and calculate Z-scores (WAZ, LAZ/HAZ, WLZ/WHZ, HCZ, BMI-for-age > 5 y)	Underweight, stunting, wasting, micro/macrocephaly; trend
Growth velocity (cm/year over ≥ 6 months)	< 4 cm/year in a pre-pubertal child is abnormal
Mid-parental height	Boys (father + mother + 13)/2; girls (father + mother − 13)/2; range ± 8.5 cm; is the child within the genetic target?
Glucose	Hypoglycaemia in sick or malnourished children
HIV rapid test (DNA PCR < 18 months if exposed)	HIV
Hb (HemoCue)	Anaemia (iron deficiency, malaria, hookworm)
Urine dipstick	UTI, glucose (diabetes), protein (renal disease)
Simple hearing and vision tests	Distraction test, response to voice; fixing, following, red reflex
Developmental screening tool	Milestones checklist; Ages and Stages Questionnaire (ASQ-3), **Malawi Developmental Assessment Tool (MDAT)**, Guide for Monitoring Child Development (GMCD), Denver II

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
FBC, ferritin (if available), blood film	Pallor, faltering, poor diet	Microcytic anaemia; malaria parasites	Iron, antimalarial, deworming. HC IV and above (ferritin at RRH/NRH)
HIV test (DNA PCR if < 18 months)	Faltering, delay, recurrent infections, unknown status	Positive	ART, co-trimoxazole (all levels; PCR via hub system)

Test	When	Expected / abnormal finding	How it changes management
TB workup (CXR, Xpert Ultra on sputum/stool/gastric aspirate)	Cough > 2 weeks, contact, weight loss, faltering despite feeding	Xpert positive, hilar nodes	TB treatment (WHO 2022: 4-month regimen for non-severe TB). Xpert at HC IV and above; stool Xpert increasingly available
Urine culture, creatinine, electrolytes	Faltering without cause, polyuria, abnormal dipstick	CKD, renal tubular acidosis, UTI	Specific treatment. RRH/NRH
TSH, free T4	Short stature with falling height velocity, delayed bone age, delay, constipation, prolonged jaundice	High TSH, low T4	Levothyroxine. RRH/NRH/private
Bone age (left hand and wrist X-ray)	Short stature, delayed or precocious puberty	Delayed (constitutional delay, hypothyroidism, GH deficiency, chronic disease); advanced (precocious puberty, obesity, CAH)	Differentiates normal variants from pathology. Hospitals with X-ray
Wrist/knee X-ray, ALP, calcium, phosphate	Signs of rickets	Cupping, fraying, widened metaphyses; raised ALP	Vitamin D and calcium
Karyotype	Short girl (any unexplained short stature: Turner), dysmorphism, global delay	45,X; trisomy 21	Genetic counselling, specific care. Mulago/MakCHS genetics or private labs; costly
Coeliac serology, stool microscopy	Chronic diarrhoea, faltering	Positive tTG; parasites (Giardia, hookworm)	Diet; antiparasitic treatment
Neuroimaging (USS, CT, MRI), EEG	Developmental delay with abnormal head size, neurological signs, seizures, regression	Hydrocephalus, malformation, HIE pattern, atrophy	Neurosurgery, counselling, prognosis. CT at RRH/NRH; MRI Kampala
Formal hearing test (OAE, ABR) and ophthalmology	Any speech delay; at-risk babies (prematurity, meningitis, kernicterus, CMV)	Hearing loss; visual impairment	Hearing aids, glasses, communication plans. Mulago, CoRSU, some RRHs
GH stimulation tests, LH/FSH, sex steroids	Specialist evaluation of short stature/puberty	GH deficiency; central vs peripheral precocious puberty	Specialist endocrine care (Mulago)

Step 5: Putting it together

Model one-line summary

> "A 9-month-old girl, born at term weighing 3.1 kg, exclusively breastfed to 4 months then fed thin maize porridge twice a day, with recurrent diarrhoea, now weighing 6.6 kg (WAZ –2.0), length 66 cm (LAZ –1.7), WLZ about –1.2, MUAC 12.3 cm, HC 43 cm (normal), whose weight has crossed down two centile spaces since 4 months, with age-appropriate milestones (sits without support, transfers, babbles 'mamama', stranger anxiety): growth faltering with moderate underweight and MAM by MUAC due to inadequate complementary feeding and recurrent diarrhoea; development normal."

Problem list

1. Growth faltering, MAM by MUAC (12.3 cm), moderate underweight.
2. Inadequate complementary feeding (thin, infrequent, low-diversity diet).

3. Recurrent diarrhoea.
4. HIV status unknown (test).
5. Normal development.

EXAM FAVOURITE

Worked example 1: plotting and interpreting. A boy aged 2 years 6 months: weight 10.2 kg, height 84 cm. On WHO charts: WAZ ≈ −2.6 (underweight), HAZ ≈ −2.4 (**stunted**), WHZ ≈ −1.3 (not wasted). His weight-for-height is acceptable because he is short and light **in proportion**. Interpretation: **chronic undernutrition (stunting)** rather than acute wasting; look for long-standing poor diet, recurrent infections, HIV, TB, and family height; counsel on complementary feeding and follow growth velocity. He does not meet SAM or MAM criteria.

EXAM FAVOURITE

Worked example 2: mid-parental height. A 12-year-old girl is 136 cm (HAZ ≈ −2.1). Father 165 cm, mother 152 cm. MPH = (165 + 152 − 13) ÷ 2 = **152 cm**; target range **143.5–160.5 cm**. Her centile is consistent with her parents' (MPH ≈ −1.5 to −2 SD on adult charts), her growth velocity over a year is 5.5 cm, she is B2 and her bone age is 11 years. Interpretation: **familial short stature with mild constitutional delay**; reassure and follow velocity. If she were B1 at 13, crossing centiles, or had a velocity < 4 cm/year, investigate (TSH, karyotype for Turner syndrome, coeliac, chronic disease).

Differential diagnosis

Causes of growth faltering and short stature

Group	Examples	Clues
Inadequate intake	Poor complementary feeding, food insecurity, breastfeeding problems, neglect	Diet history; catch-up with feeding
Infection	HIV, TB, recurrent diarrhoea, malaria, intestinal parasites	Tests; chronic symptoms
Malabsorption	Coeliac disease, persistent diarrhoea, cystic fibrosis (rare)	Chronic diarrhoea, bulky stools
Chronic organ disease	Congenital heart disease, CKD, sickle cell disease, chronic lung disease, liver disease	Murmur, failure to thrive with breathlessness, anaemia
Endocrine	Hypothyroidism, GH deficiency, Cushing's/steroid use, rickets	Height affected more than weight; bone age delayed
Genetic / syndromic	Down syndrome, Turner syndrome, skeletal dysplasias (achondroplasia), SGA without catch-up	Dysmorphism, disproportion, use syndrome-specific charts
Normal variants	Familial short stature, constitutional delay	Normal velocity; MPH; bone age
Psychosocial	Emotional deprivation, abuse	Improves in a caring environment

Causes of developmental delay

Group	Examples
Prenatal	Genetic (Down syndrome, fragile X), congenital infections (CMV, rubella, toxoplasmosis, syphilis, HIV), fetal alcohol spectrum disorder, brain malformations, hypothyroidism
Perinatal	Prematurity, **birth asphyxia/HIE**, **kernicterus**, neonatal hypoglycaemia, neonatal sepsis/meningitis

Group	Examples
Postnatal	**Cerebral malaria**, **meningitis**, encephalitis, head injury, severe malnutrition, iodine/iron deficiency, lead poisoning, epilepsy
Sensory	Hearing loss, visual impairment
Neurodevelopmental	Autism spectrum disorder, specific language disorder
Neuromuscular	Duchenne muscular dystrophy, spinal muscular atrophy
Environmental	Lack of stimulation, neglect, maternal depression
Progressive (regression)	HIV encephalopathy, metabolic and neurodegenerative disorders, Rett syndrome, SSPE after measles

Investigations

Test	What you are looking for	Why
Accurate measurement and plotting	Z-scores and trend	Diagnoses faltering, stunting, wasting, abnormal head growth
HIV test	HIV infection	Common, treatable cause of faltering and delay (UCG 2023)
Hb / FBC	Anaemia	Affects growth and development
TB screen	TB	Chronic weight loss
TSH	Hypothyroidism	Treatable cause of short stature and delay
Bone age	Skeletal maturity	Separates constitutional delay from pathology
Karyotype	Turner, Down syndrome	Short girls; dysmorphism
Hearing and vision tests	Sensory deficits	Speech delay; visual delay
Neuroimaging/EEG	Brain lesion, seizures	Global delay with neurological signs

Management

A. Growth monitoring and promotion (every child, every contact)

1. **Weigh, measure length/height and MUAC** (from 6 months); **HC** in under-2s and in any child with delay.
2. **Plot** on the WHO chart / Child Health Card; calculate Z-scores; **compare with the last point**.
3. **Classify**: normal, growth faltering, MAM, SAM, stunting, overweight.
4. **Counsel the caregiver on the trend**: praise good growth; explore causes of poor growth.
5. **Act** (see below) and book follow-up.

B. Managing growth faltering and undernutrition step by step

1. **Exclude SAM** (WHZ < −3, MUAC < 11.5 cm or oedema) → SAM protocol (see the malnutrition topic): **outpatient therapeutic care (OTC)** with RUTF if appetite good and no complications; **inpatient (ITC)** with F-75 then F-100/RUTF if complicated. WHO 2023 guideline on wasting also covers infants < 6 months at risk of poor growth and development.
2. **MAM**: nutrition counselling and supplementary feeding where available (WHO 2023 suggests specially formulated foods for selected children), treat illness, review every 2 weeks.

3. **Treat underlying disease**: HIV (ART), TB, malaria, diarrhoea (ORS and zinc), worms, UTI, anaemia.
4. **Feeding counselling** (WHO 2023 guideline on complementary feeding of children 6–23 months; IMCI):
    - **Exclusive breastfeeding** for 6 months; continue breastfeeding to 2 years or beyond.
    - **Complementary foods from 6 months**: thick enough to stay on a spoon; **2–3 meals/day at 6–8 months**, **3–4 meals/day at 9–23 months** plus 1–2 snacks.
    - **Dietary diversity**: at least **5 of 8 food groups** daily; **animal-source foods daily** (eggs, milk, fish, meat, insects), legumes and nuts (beans, groundnut/simsim paste), vitamin A–rich fruits and vegetables (pumpkin, mango, dark-green leaves, orange-fleshed sweet potato).
    - **Responsive feeding**: feed slowly and patiently, encourage, use a separate plate; continue feeding during illness and give extra afterwards.
    - **Hygiene**: safe water, handwashing, clean utensils.
5. **Micronutrients and deworming** (UCG 2023; table below).
6. **Follow-up**: monthly weight; refer if no improvement after 2–3 months, if height crosses down, or if red flags appear.

Drug / supplement	Dose	Route	Frequency	Duration / max	Notes
Vitamin A	**< 6 months: 50,000 IU**; **6–11 months: 100,000 IU**; **12–59 months: 200,000 IU**	Oral	**Every 6 months** from 6 months (routine)	Treatment: 3 doses (day 0, 1 and 4 weeks) for measles, persistent diarrhoea or eye signs (UCG); in SAM follow the SAM protocol	Do not repeat within 1 month (toxicity) except as a treatment schedule
Albendazole	**1–2 years: 200 mg**; **> 2 years: 400 mg**	Oral	Single dose **every 6 months**		Or mebendazole 250 mg (1–2 y) / 500 mg (> 2 y) single dose (UCG 2023)
Iron (ferrous sulphate, elemental iron)	Treatment of iron-deficiency anaemia: **3 mg/kg/day** elemental iron (range 3–6); UCG HC band: syrup 25 mg/mL, 1 mL daily < 1 year; ½ tablet (200 mg) daily 1–5 years	Oral, after meals	Once daily	14 days in IMCI, then reassess; continue ~3 months after Hb normalises in confirmed deficiency	Treat malaria first in malarious areas; deworm; **do not give iron in the acute phase of SAM**
Zinc (for diarrhoea)	**2–6 months: 10 mg; > 6 months: 20 mg**	Oral	Once daily	**10 days** (UCG 2023)	With ORS; reduces duration and recurrence
Vitamin D (rickets)	Treatment 2,000–6,000 IU/day (or a single high dose per specialist) + calcium 500 mg/day; prevention 400 IU/day	Oral	Daily	2–3 months then maintenance	Check ALP and X-ray response

C. Managing developmental concerns step by step

1. **Confirm** the delay with a structured assessment (use corrected age if preterm); classify: single domain vs **global**; delay vs **regression** (regression = urgent specialist referral).
2. **Check hearing and vision** in every child with delay (especially speech delay).
3. **Look for a cause**: history, examination (head size, dysmorphism, neurocutaneous signs, neurology), **HIV test** (UCG 2023), Hb, TSH; imaging, karyotype or EEG as indicated.
4. **Treat what is treatable**: HIV (ART), hypothyroidism (levothyroxine), malnutrition, anaemia, epilepsy, hearing loss (hearing aids, grommets), visual problems (glasses, cataract surgery).
5. **Early intervention** (do not wait for a diagnosis): physiotherapy, occupational therapy, speech and language therapy; **teach parents to play, talk, read and sing** with the child (WHO/UNICEF **Care for Child**

Development), responsive caregiving; community-based rehabilitation.

6. **Support the family**: explain clearly, address stigma and beliefs, screen the mother for depression, link to parent groups, social support, **inclusive education** and later vocational training.
7. **Review** every 3 months; re-assess milestones; re-plot growth.

D. Puberty problems (what to do on the ward or in clinic)

Problem	First steps	Refer
Precocious puberty (girls < 8, boys < 9)	Tanner stage, growth velocity, bone age, neuro exam (headache, visual fields), abdominal/testicular exam, café-au-lait spots (McCune–Albright)	**All boys** and girls < 6–7 years or with rapid progression/neurological signs → paediatric endocrinology (Mulago) for LH/FSH, sex steroids, pelvic USS, MRI brain
Delayed puberty (girls no breasts at 13; boys testes < 4 mL at 14)	Growth chart, MPH, family history, chronic disease screen (HIV, sickle, TB, renal, IBD), nutrition, smell (Kallmann), bone age	Short girl → **karyotype** (Turner); persistent delay → LH/FSH, specialist
Premature thelarche (isolated breast buds, girls < 2 years, no growth spurt)	Reassure, follow-up	If progression or growth acceleration

Complications

Of poor growth	Of untreated developmental delay
Increased risk of death from infection (wasting and stunting together multiply risk)	Poor school performance, exclusion
Impaired cognitive development, lower school attainment and adult earnings (stunting)	Behavioural problems, low self-esteem
Short adult stature; obstructed labour in short women (next generation)	Secondary problems (contractures, malnutrition in CP)
Later obesity and non-communicable disease after rapid catch-up	Family stress, maternal depression

Ward-round & exam pearls

Model presentation

> "Nabukeera is a 9-month-old girl from Mpigi, born at term weighing 3.1 kg, HIV-exposed (mother on ART since pregnancy, virally suppressed), with a negative DNA PCR at 6 weeks and on co-trimoxazole; immunised up to date (BCG, OPV, IPV, DPT-HepB-Hib, PCV, rotavirus; MR due today). Vitamin A and deworming not yet given. She was exclusively breastfed to 4 months, then given thin maize porridge twice a day plus breast milk; no animal-source foods. She has had 3 episodes of diarrhoea in 2 months. Weight 6.6 kg (WAZ −2.0), length 66 cm (LAZ −1.7), WLZ about −1.2, MUAC 12.3 cm (MAM), HC 43 cm (normal), no oedema; weight has crossed down from about the 25th to below the 3rd centile since 4 months. Development: sits without support, transfers objects, rakes small items, babbles 'mamama', shows stranger anxiety: **normal for age**. **Impression: growth faltering with moderate acute malnutrition by MUAC, due to inadequate complementary feeding and recurrent diarrhoea; development normal.** Plan: repeat HIV test per PMTCT schedule (DNA PCR at 9 months per national algorithm), give MR vaccine, vitamin A 100,000 IU, counsel on 3–4 thick meals a day with eggs, beans and groundnut paste plus continued breastfeeding, handwashing, zinc and ORS for future diarrhoea, review in 2 weeks."

Viva questions (with answers)

1. **What is a Z-score?** The number of SDs a measurement lies from the WHO median for age and sex; < −2 moderate, < −3 severe.
2. **Difference between stunting and wasting?** Stunting = low height-for-age (chronic); wasting = low weight-for-height or low MUAC (acute).
3. **SAM criteria?** WHZ < −3, MUAC < 11.5 cm (6–59 months), or bilateral pitting oedema.
4. **When should a child sit, walk and talk?** Sit by 9 months, walk by 18 months, 6 words by 18 months, two-word phrases by 2 years (limit ages).
5. **Name the domains of development.** Gross motor; fine motor and vision; speech, language and hearing; social, emotional and self-care.
6. **What is global developmental delay?** Significant delay in two or more domains in a child under 5.
7. **When does the anterior fontanelle close?** 12–18 months.
8. **Define precocious and delayed puberty.** Precocious: girls < 8, boys < 9 years. Delayed: no breast development by 13 (girls), testes < 4 mL by 14 (boys).
9. **How do you calculate mid-parental height?** Boys (F + M + 13)/2; girls (F + M − 13)/2, ± 8.5 cm.
10. **When do you correct for prematurity?** Until 2 years of age.

OSCE tips

- **Developmental assessment station**: introduce yourself, explain to the parent, **observe the child playing first**, then go domain by domain (gross motor, fine motor/vision, language/hearing, social), using simple toys (cubes, small bean or raisin under supervision, ball, crayon and paper, picture book, rattle). State the **developmental age in each domain** and compare with the chronological (corrected) age; finish with "I would also check hearing and vision, plot growth and head circumference."
- **Growth chart station**: check the chart matches the child's **sex and age range** and the **measurement type**; plot accurately (age on x-axis, measurement on y-axis); comment on the **trend**, Z-score and proportionality; give an interpretation and a plan.

Mnemonics & memory aids

- **Weight rule "2–1–2.5"**: **doubles** by 5 months, **triples** by **1** year, **quadruples** by **2.5** years.
- **Length "50–75–100"**: 50 cm at birth, 75 cm at 1 year, 100 cm at 4 years.
- **HC "35–46"**: 35 cm at birth, about 46 cm at 1 year (2–1–0.5 cm/month per quarter-year).
- **Limit ages "Smile 2, See 3, Head 4, Reach 6, Sit 9, Pinch 12, Walk & Words 18, Phrases 2"**: smile by 2 months, fix and follow by 3, head control by 4, reach by 6, sit by 9, pincer by 12 months, walk and 6 words by 18 months, two-word phrases by 2 years.
- **Drawing shapes "circle 3, cross 4, square 4.5, triangle 5"** (the number of years matches the age).
- **Cubes**: tower of 3 at 18 months, 6–7 at 2 years, 9 at 3 years (roughly age in years × 3).
- **Puberty order, girls "Buds, Bush, Growth, Gush"**: thelarche → pubarche → growth spurt → menarche. **Boys "Balls first"**: testes ≥ 4 mL.

High-yield summary

Numbers & doses to know

Item	Value
Birthweight loss	Up to **10%** in the first week; regained by **2–3 weeks**

Item	Value
Weight	Doubles **4–5 months**; triples **12 months**; quadruples **~2.5 years**
Length	**50 → 75 → 87 → 100 cm** (birth, 1, 2, 4 years); then **5–6 cm/year**
Head circumference	**35 cm → ~46 cm** at 1 year
Anterior fontanelle	Closes **12–18 months** (posterior by 6–8 weeks)
First tooth	**~6 months**; 20 teeth by 2.5–3 years
Z-score cut-offs	**< −2** moderate; **< −3** severe; **> +2** overweight (WHZ, < 5 y); **> +3** obese
SAM	WHZ < −3, **MUAC < 11.5 cm**, or **bilateral pitting oedema**
MAM	WHZ −3 to −2, **MUAC 11.5–12.4 cm**
Length vs height	Lying < 2 y (or < 87 cm); standing ≥ 2 y; subtract 0.7 cm if ≥ 2 y measured lying
Limit ages	Smile **8 weeks**; fix/follow **3 months**; head control **4 months**; reach **6 months**; sit **9 months**; pincer **12 months**; walk **18 months**; 6 words **18 months**; 2-word phrases **2 years**
Primitive reflexes	Gone by **3–6 months**; parachute from **6–9 months**
Global developmental delay	≥ **2 domains**, child < 5 years; DQ < 70
Corrected age	Until **2 years**
Precocious puberty	Girls **< 8 y**, boys **< 9 y**
Delayed puberty	Girls no breasts by **13 y**; boys testes < 4 mL by **14 y**
Menarche	**~2–2.5 years after thelarche**, at B4
Mid-parental height	Boys (F + M + 13)/2; girls (F + M − 13)/2; **± 8.5 cm**
Vitamin A	**50,000 IU** (< 6 m), **100,000 IU** (6–11 m), **200,000 IU** (12–59 m) every 6 months
Albendazole	**200 mg** (1–2 y), **400 mg** (> 2 y) every 6 months
Zinc (diarrhoea)	**10 mg** (2–6 m), **20 mg** (> 6 m) daily for **10 days**

Classic exam traps

- **"Weight is normal, so nutrition is fine."** Wrong: check **oedema** and **height**; stunting and kwashiorkor can coexist with "normal" weight.
- **"Measure MUAC in a 3-month-old."** Wrong: MUAC cut-offs apply from **6 months** (UCG; WHO 6–59 months).
- **"A 10-month-old ex-28-week baby who cannot sit has motor delay."** Wrong: corrected age is ~7 months; sitting is not yet due.
- **"Hand preference at 9 months shows she will be right-handed."** Wrong: it is abnormal before 12 months; examine the other arm.
- **"Isolated speech delay is laziness; wait and see."** Wrong: test **hearing** first.
- **"Short child = growth hormone deficiency."** Wrong: commonest causes are familial short stature, constitutional delay and undernutrition; always plot the parents and the velocity.
- **"Precocious puberty in a boy is usually idiopathic."** Wrong: it is often pathological; refer all boys.
- **"Regression in a toddler is just a phase."** Wrong: loss of skills is always a red flag (HIV encephalopathy, metabolic, Rett syndrome, autism).

Questions seniors ask

How do you tell stunting from wasting, and which is worse? Stunting is low height-for-age (chronic, mostly from the first 1000 days); wasting is low weight-for-height/MUAC (acute). Wasting carries the higher immediate risk of death; stunting has lifelong effects on cognition and adult health; children with both have the highest mortality.

This child's weight has crossed two centile lines. What do you do? Re-measure carefully, take a feeding and illness history, test for HIV, screen for TB, malaria, UTI and anaemia, examine for chronic disease, counsel on feeding and review in 2–4 weeks; treat SAM/MAM if criteria are met.

Why is head circumference important? It reflects brain growth; microcephaly (or a falling HC) suggests brain injury, congenital infection, genetic disease or HIV encephalopathy; macrocephaly or a rising HC suggests hydrocephalus or a mass.

How do you assess development quickly on a ward round? Ask the parent what worries them and what the child does now; observe the child; check one or two skills per domain against the limit ages; ask about regression; note hearing and vision.

A 2-year-old says no words but walks and plays well. What do you think of? Hearing loss first (test), then autism spectrum disorder (eye contact, pointing, response to name), global delay, lack of stimulation, or a specific language disorder.

What is the difference between familial short stature and constitutional delay? Familial: short parents, normal velocity, normal bone age, normal puberty timing, short final height. Constitutional: often average parental heights with a family history of late puberty, normal velocity, **delayed bone age** and delayed puberty, normal final height.

What advice would you give the mother of a 7-month-old on complementary feeding? Continue breastfeeding; give thick porridge enriched with groundnut/simsim paste, milk or egg, mashed beans, fish, vegetables and fruit; 2–3 meals a day now and 3–4 plus snacks from 9 months; feed patiently with a separate bowl; wash hands; continue feeding during illness.

SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Four domains of development?	**Gross motor**; **fine motor and vision**; **speech, language and hearing**; **social, emotional and self-care** (cognition is often added).
2	When does birthweight double, triple and quadruple?	Doubles by **4–5 months**, triples by **12 months**, quadruples by about **2.5 years**.
3	Normal weight loss after birth and when birthweight is regained?	Up to **7–10%** in the first week; back to birthweight by **2–3 weeks** (NICE NG75: most by 3 weeks; > 10% loss needs feeding assessment).
4	Length at birth, 1 year, 2 years and 4 years?	**50 cm → 75 cm → ~87 cm → ~100 cm**; then about **5–6 cm/year** until puberty.
5	Head circumference at birth and 1 year, and the monthly increments?	**35 cm → ~46 cm**: +2 cm/month (0–3 m), +1 cm/month (4–6 m), +0.5 cm/month (7–12 m).
6	Define a Z-score and the cut-offs used.	SDs from the WHO median for age and sex. **< −2 moderate, < −3 severe**; WHZ > +2 overweight, > +3 obese (under 5).
7	Stunting vs wasting vs underweight?	**Stunting** = low height-for-age (chronic); **wasting** = low weight-for-height or MUAC (acute); **underweight** = low weight-for-age (composite).
8	SAM criteria (WHO 2023, UCG 2023)?	**WHZ < −3**, or **MUAC < 11.5 cm** (6–59 months), or **bilateral pitting oedema**.
9	MAM criteria?	**WHZ −3 to < −2** or **MUAC 11.5–12.4 cm**.
10	How do you measure MUAC correctly?	**Left** arm, at the **midpoint** between acromion and olecranon, arm hanging relaxed, tape snug but not compressing; only from **6 months**.
11	Length or height: which, when?	**Length lying** under 2 years (or < 87 cm), with two people on a length board; **height standing** from 2 years. Subtract 0.7 cm if a ≥ 2-year-old is measured lying.
12	Gross motor limit ages?	Head control **4 months**; sits unsupported **9 months**; walks alone **18 months**.
13	Fine motor and vision limit ages?	Fixes and follows **3 months**; reaches **6 months**; transfers **9 months**; pincer grip **12 months**.
14	Speech and language limit ages?	Babble by 7–10 months; **6 words with meaning by 18 months**; **2-word phrases by 2 years** (2.5 years at the latest).
15	Social limit ages?	**Social smile by 8 weeks**; stranger anxiety by 10 months; feeds self with spoon by 18 months; symbolic play by 2–2.5 years.
16	Shapes a child can copy by age?	**Circle 3 y, cross 4 y, square 4.5 y, triangle 5 y**; vertical line 2 y.
17	Cube towers by age?	**3 cubes at 18 months**, **6–7 at 2 years**, **9 at 3 years**.
18	When should the Moro, palmar grasp and ATNR disappear, and when does the parachute appear?	Moro **3–6 months**, palmar grasp **3–4 months**, ATNR **4–6 months**; **parachute appears 6–9 months** and persists.
19	Define global developmental delay.	Significant delay in **two or more domains** in a child **under 5 years** (DQ < 70).
20	How long do you correct for prematurity?	Until **2 years**: corrected age = chronological age − weeks born before 40 weeks.

#	Question	Answer
21	When does the anterior fontanelle close, and causes of delayed closure?	**12–18 months** (up to 24). Late: **rickets, hypothyroidism, hydrocephalus, Down syndrome**, osteogenesis imperfecta.
22	First sign of puberty in girls and boys?	Girls: **breast bud (thelarche, B2)** ~10–11 y. Boys: **testicular volume ≥ 4 mL (G2)** ~11–12 y.
23	Definitions of precocious and delayed puberty?	Precocious: girls **< 8 y**, boys **< 9 y**. Delayed: girls no breasts by **13 y**; boys testes < 4 mL by **14 y**.
24	When does menarche occur relative to thelarche, and peak height velocity in each sex?	Menarche ~**2–2.5 years after thelarche**, at **B4**. PHV: girls early (B2–3, ~12 y); boys late (G4, ~14 y).
25	Mid-parental height formula?	Boys: **(father + mother + 13) ÷ 2**; girls: **(father + mother − 13) ÷ 2**; target range **± 8.5 cm**.
26	NICE NG75 thresholds for faltering growth?	Fall across **≥ 1** centile space if birthweight < 9th; **≥ 2** if 9th–91st; **≥ 3** if > 91st; or weight **< 2nd centile**.
27	Routine vitamin A and albendazole doses (UCG 2023)?	Vitamin A **100,000 IU (6–11 m)**, **200,000 IU (12–59 m)** every 6 months. Albendazole **200 mg (1–2 y)**, **400 mg (> 2 y)** every 6 months.
28	Why must a child with developmental delay or growth faltering in Uganda have an HIV test?	HIV causes failure to thrive (up to 50% of infected children) and encephalopathy with delay/regression; **ART** reverses or halts it (UCG 2023).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 10-month-old boy born at 32 weeks' gestation is brought for review. He rolls both ways and sits with support but not alone; he reaches and transfers objects, babbles 'baba', and smiles at his mother. Tone and reflexes are normal and his head circumference follows the 25th centile. What is the MOST appropriate interpretation?

- A Global developmental delay: refer for MRI
- B Gross motor delay suggesting cerebral palsy
- C Normal development for chronological age
- D Normal development for his corrected age of about 8 months; review in 2 months
- E Hearing loss causing delay

2. A 2-year-6-month-old boy weighs 10.2 kg and is 84 cm tall. On WHO charts his WAZ is −2.6, HAZ −2.4 and WHZ −1.3. MUAC is 13.4 cm with no oedema. How should his nutritional status be classified?

- A Severe acute malnutrition
- B Stunted and underweight, not wasted
- C Moderate acute malnutrition
- D Normal nutritional status
- E Severe stunting with wasting

3. A 9-month-old girl is brought by her mother, who is worried she always reaches with her right hand and 'keeps the left hand fisted'. She sits alone and babbles. What is the MOST appropriate next step?

- A Reassure: she will be right-handed
- B Advise the mother to encourage the left hand and review at 2 years
- C Order a karyotype
- D Prescribe vitamin D
- E Examine carefully for left-sided weakness and refer for neurological assessment

4. A 15-month-old boy (8.0 kg) has bilateral pitting oedema of both feet and lower legs, MUAC 12.0 cm, WHZ −1.8. He eats the RUTF appetite test well and has no danger signs. What is the MOST appropriate classification and plan according to WHO 2023/UCG 2023? Advanced

- A SAM (oedema); outpatient therapeutic care with RUTF if no complications and appetite is good
- B MAM; supplementary feeding only
- C Normal nutrition because the WHZ is above −2
- D Nephrotic syndrome; start prednisolone
- E Admit and give IV furosemide for the oedema

5. A 2-year-old girl walks, runs, builds a tower of 6 cubes and feeds herself, but says no words. She does not turn when her name is called from behind and had meningitis at 8 months. What is the MOST appropriate next step?

- A Reassure: girls often speak late
- B Diagnose autism spectrum disorder and refer to a psychiatrist
- C Formal hearing assessment (OAE/ABR) and audiology referral
- D Start speech therapy only
- E MRI brain

6. Which pair of milestones would a typical 12-month-old child be EXPECTED to show?

A Copies a circle and rides a tricycle
B Builds a tower of 6 cubes and uses 2-word phrases
C Hops on one foot and dresses without help
D Draws a person with 6 parts and skips
E Neat pincer grip and walks holding on to furniture

7. A 7-month-old boy's head circumference was on the 50th centile at birth and at 3 months, but now it is above the 98th centile. He is irritable and vomits; the fontanelle is tense and he looks downwards with the sclera visible above the iris. What is the MOST appropriate next step?

A Urgent cranial ultrasound through the fontanelle (or CT) and neurosurgical referral
B Reassure: large heads run in families
C Measure again in 3 months
D Start oral vitamin D for rickets
E Lumbar puncture

8. A 14-year-old boy is the shortest in his class (HAZ −2.0). His height velocity over the last year is 5 cm. Testicular volume is 3 mL. His father 'grew late, at 17'. Bone age is 11.5 years. Examination is otherwise normal and screening tests (Hb, HIV, urine, TSH) are normal. What is the MOST likely diagnosis? Advanced

A Growth hormone deficiency
B Klinefelter syndrome
C Hypothyroidism
D Constitutional delay of growth and puberty
E Familial short stature

9. A mother brings her 18-month-old son because 'he used to say mama, dada and ball and wave bye-bye, but over 3 months he has stopped talking, does not look at us, and flaps his hands'. He walks well. His HIV test is negative and hearing is normal. What is the MOST important feature in deciding the next step?

A His walking is normal, so his development is reassuring
B Boys often talk late, so review at 3 years
C Loss of previously acquired skills (regression), requiring urgent specialist referral
D He needs vitamin A
E He needs a growth chart only

10. A 6-year-old girl has breast budding (B2) and pubic hair (PH2), with a height velocity of 9 cm in the last year. Her bone age is 8.5 years. What is the MOST appropriate classification and action?

A Normal puberty: reassure
B Precocious puberty: refer to paediatric endocrinology for evaluation (LH/FSH, pelvic ultrasound, consider MRI brain)
C Premature thelarche: reassure and review in a year
D Delayed puberty
E Obesity-related pseudo-puberty: dietary advice only

11. A 7-month-old girl is weighed at a HC III. Her weight has fallen from the 50th centile at 4 months to the 9th centile now (crossing 2 centile spaces). She is breastfed and given thin porridge once daily. MUAC 12.8 cm, no oedema, WHZ −1.4, development normal. Her mother's HIV status is unknown. What is the BEST plan?

A Test mother and child for HIV, counsel on complementary feeding (thick, diverse meals 2–3 times daily plus breastfeeding), give vitamin A 100,000 IU, and review in 2–4 weeks
B Admit for F-75 therapeutic milk
C Start RUTF for SAM
D Stop breastfeeding and start formula
E No action: she is not wasted

12. A 4-year-old boy cannot copy a circle, speaks only single words, does not play with other children, and cannot walk up stairs. He has a history of cerebral malaria at 18 months with prolonged coma. Hearing and vision are normal. Which term BEST describes his development?

A Isolated speech delay
B Constitutional delay
C Global developmental delay
D Autism spectrum disorder
E Normal development

13. You are measuring a 20-month-old girl on the ward. Which technique is CORRECT?

A Measure standing height against a wall with shoes on
B Measure MUAC on the right arm while it is flexed at 90°
C Measure head circumference around the ears and below the occiput
D Weigh her fully dressed and subtract an estimate for clothes
E Measure length lying on a length board with two people, head against the headboard and knees straightened

14. A 13-year-old girl (HAZ −2.8) has no breast development. She has a low posterior hairline, a wide carrying angle and widely spaced nipples. Her parents are of average height. What is the MOST appropriate investigation? Advanced

A Bone age only and reassure
B Karyotype
C Growth hormone stimulation test first
D Pelvic MRI
E Serum calcium

15. Which statement about normal growth and development is CORRECT?

A The posterior fontanelle closes at 12–18 months
B The palmar grasp reflex should persist until 12 months
C Boys reach peak height velocity earlier in puberty than girls
D A child's height at 2 years is roughly half of adult height
E The parachute reflex disappears by 6 months

Answers & explanations

1. D · Normal development for his corrected age of about 8 months; review in 2 months

D is right: born 8 weeks early, his **corrected age is about 8 months**; sitting alone has a limit age of 9 months and his other skills and neurology are normal. A is wrong: there is no delay in two or more domains once corrected. B is wrong: normal tone, reflexes and other skills make CP unlikely; sitting is not yet overdue. C is wrong: you must use corrected age until 2 years; by chronological age you would wrongly call him delayed. E is wrong: babbling and social smiling suggest normal hearing.

2. B · Stunted and underweight, not wasted

B is right: HAZ < −2 = **stunting** and WAZ < −2 = **underweight**, but WHZ −1.3 and MUAC 13.4 cm mean he is **not wasted** (chronic, proportionate undernutrition). A is wrong: SAM needs WHZ < −3, MUAC < 11.5 cm or oedema. C is wrong: MAM needs WHZ −3 to −2 or MUAC 11.5–12.4 cm. D is wrong: two Z-scores are below −2. E is wrong: HAZ is −2.4 (moderate, not severe < −3) and there is no wasting.

3. E · Examine carefully for left-sided weakness and refer for neurological assessment

E is right: **hand preference before 12 months is abnormal** and suggests weakness of the other side (e.g. **left hemiplegic CP**, brachial plexus injury); she needs a full neurological examination and referral. A is wrong: handedness is not established until 2–3 years. B is wrong: waiting loses time for early intervention. C is wrong: there are no dysmorphic features or global delay to suggest a chromosomal disorder. D is wrong: rickets does not cause unilateral weakness.

4. A · SAM (oedema); outpatient therapeutic care with RUTF if no complications and appetite is good

A is right: **bilateral pitting oedema = SAM** whatever the WHZ or MUAC; with good appetite and no complications he can be treated in **outpatient therapeutic care**, although oedema +++ or complications would need inpatient care. B is wrong: oedema overrides MUAC; this is not MAM. C is wrong: oedema raises weight and hides wasting. D is wrong: nephrotic syndrome must be excluded with a urine dipstick, but in this context the default is nutritional oedema; steroids are not started blindly. E is wrong: diuretics are contraindicated for nutritional oedema.

5. C · Formal hearing assessment (OAE/ABR) and audiology referral

C is right: **isolated speech delay** with no response to her name and a history of **meningitis** (a major cause of sensorineural deafness) means **hearing must be tested first**. A is wrong: no words at 2 years is beyond the 18-month limit age. B is wrong: autism is considered only after hearing is excluded; her play and self-care are good. D is wrong: therapy without knowing her hearing status will fail. E is wrong: imaging does not answer the key question.

6. E · Neat pincer grip and walks holding on to furniture

E is right: at about 12 months children have a **neat pincer grip** and **cruise** or begin to walk. A is wrong: copying a circle and riding a tricycle are 3-year skills. B is wrong: a 6-cube tower and 2-word phrases are 2-year skills. C is wrong: hopping and dressing independently are 4-year skills. D is wrong: drawing a detailed person and skipping are 5-year skills.

7. A · Urgent cranial ultrasound through the fontanelle (or CT) and neurosurgical referral

A is right: an HC **crossing centiles upward**, a tense fontanelle, vomiting and **sunsetting eyes** indicate **hydrocephalus**; image urgently (USS through the open fontanelle) and refer to neurosurgery (e.g. CURE Children's Hospital Mbale, Mulago). B is wrong: familial macrocephaly follows a high centile from birth without symptoms. C is wrong: delay risks brain damage. D is wrong: rickets can cause frontal bossing and a large fontanelle but not raised ICP. E is wrong: LP is dangerous with raised ICP and obstructive hydrocephalus.

8. D · Constitutional delay of growth and puberty

D is right: a short boy with **normal pre-pubertal velocity**, **delayed puberty** (testes < 4 mL at 14), **delayed bone age** and a **family history of late puberty** has **constitutional delay**; expect normal final height. A is wrong: GH deficiency causes a low velocity (< 4 cm/year) with falling centiles. B is wrong: Klinefelter boys are tall with small testes after puberty starts. C is wrong: TSH is normal. E is wrong: in familial short stature puberty and bone age are normal.

9. C · Loss of previously acquired skills (regression), requiring urgent specialist referral

C is right: **regression** of language and social skills is always a red flag (autism spectrum disorder with regression, epileptic encephalopathy, Rett-like or metabolic disorders) and needs urgent specialist assessment. A is wrong: normal motor skills do not offset loss of language and social skills. B is wrong: this is not late talking but loss of skills. D is wrong: vitamin A does not address regression. E is wrong: a growth chart (including HC) is part of the workup but not the deciding feature.

10. B · Precocious puberty: refer to paediatric endocrinology for evaluation (LH/FSH, pelvic ultrasound, consider MRI brain)

B is right: pubertal signs before **8 years** in a girl with **growth acceleration** and **advanced bone age** = **precocious puberty**; she needs endocrine evaluation (central vs peripheral; MRI brain if central, especially under 6–7 years). A is wrong: 6 years is too young. C is wrong: isolated premature thelarche has no pubic hair, no growth spurt and no bone age advance. D is wrong: this is early, not late. E is wrong: there is no mention of obesity, and true pubertal signs with rapid growth need investigation.

11. A · Test mother and child for HIV, counsel on complementary feeding (thick, diverse meals 2–3 times daily plus breastfeeding), give vitamin A 100,000 IU, and review in 2–4 weeks

A is right: crossing 2 centile spaces is **faltering growth** (NICE NG75); the likely cause is inadequate complementary feeding, but **HIV must be tested** (unknown maternal status). Feeding counselling, vitamin A and early review are appropriate. B is wrong: F-75 is for complicated SAM in inpatient care. C is wrong: she does not meet SAM criteria. D is wrong: breastfeeding should continue to 2 years and beyond. E is wrong: faltering needs action before wasting develops.

12. C · Global developmental delay

C is right: he is delayed in **fine motor, language, social and gross motor** domains (two or more) under 5 years: **global developmental delay**, here probably due to brain injury from cerebral malaria. A is wrong: more than one domain is affected. B is wrong: constitutional delay refers to growth and puberty. D is wrong: autism is defined by social communication difficulties and restricted behaviours, not by motor delay; it is not the best summary of delay in all domains after brain injury. E is wrong: he is far behind the expected milestones.

13. E · Measure length lying on a length board with two people, head against the headboard and knees straightened

E is right: under 2 years, measure **length lying** on a length board with **two people**. A is wrong: standing height is used from 2 years and never with shoes. B is wrong: MUAC is measured on the **left** arm hanging relaxed (flex only to find the midpoint). C is wrong: HC is the maximal **occipitofrontal** circumference above the eyebrows and around the occiput. D is wrong: infants should be weighed naked or in minimal clothing; estimates add error.

14. B · Karyotype

B is right: short stature, **delayed puberty** and features such as a low hairline, cubitus valgus and a broad chest suggest **Turner syndrome (45,X)**; a **karyotype** confirms it (and should be considered in any unexplained short girl). A is wrong: reassurance misses Turner syndrome with ovarian failure and cardiac/renal associations. C is wrong: GH testing is not first-line here; Turner girls are not GH-deficient. D is wrong: imaging the ovaries is secondary. E is wrong: calcium does not explain this picture.

15. D · A child's height at 2 years is roughly half of adult height

D is right: height at 2 years is roughly **half of final adult height**. A is wrong: the **posterior** fontanelle closes by 6–8 weeks; it is the anterior that closes at 12–18 months. B is wrong: the palmar grasp disappears by 3–4 months to allow voluntary reaching. C is wrong: girls reach PHV early (B2–3) and boys late (G4). E is wrong: the parachute **appears** at 6–9 months and persists for life.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The baby who stopped growing

Nabukeera, a 9-month-old girl, is brought to the Mengo young child clinic for measles-rubella vaccination. The nurse notes her weight is 6.6 kg; at 4 months it was 6.4 kg. Her mother is on ART. Birthweight was 3.1 kg at term.

1. What measurements will you take and how?

> **Weight** naked on a zeroed scale; **length lying** on a length board with two people; **MUAC** on the left arm (from 6 months); **head circumference**; check **oedema** on both feet. Plot on WHO charts for a girl and calculate WAZ, LAZ, WLZ, HCZ.

2. Results: WAZ −2.0, LAZ −1.7, WLZ −1.2, MUAC 12.3 cm, HC normal, no oedema. Interpret.

> **Growth faltering** (weight crossed from about the 25th centile to below the 3rd: ≥ 2 centile spaces) with **moderate underweight** and **MAM by MUAC** (11.5–12.4 cm); not severely wasted; length is relatively preserved, so this is recent.

3. What history will you take?

> Breastfeeding frequency; when complementary foods started; **24-hour recall** (what, how thick, how often, animal-source foods, who feeds); diarrhoea, cough, fevers; HIV exposure: mother's viral load, child's **DNA PCR results**, co-trimoxazole; immunisations, vitamin A, deworming; development; social situation and food security.

4. She was exclusively breastfed to 4 months, then given thin maize porridge twice a day, no eggs, fish or beans; 3 diarrhoeal episodes in 2 months. Development is normal. What investigations?

> **HIV DNA PCR** per the national PMTCT schedule, Hb, malaria test if febrile, urine dipstick if unexplained fevers, stool if persistent diarrhoea; TB screen if cough or contact. No complex tests needed if she responds to feeding.

5. What is your management?

> **Complementary feeding counselling**: continue breastfeeding; thick porridge enriched with groundnut/simsim paste, milk or egg; mashed beans, fish, vegetables, fruit; **3–4 meals/day from 9 months** plus snacks; responsive feeding, separate bowl. **Vitamin A 100,000 IU**, MR vaccine today, hygiene and handwashing, ORS and **zinc 20 mg daily for 10 days** for future diarrhoea; supplementary food if available. Review in **2 weeks**.

6. At review 1 month later she has lost weight and has persistent diarrhoea. What now?

> Re-measure; screen for SAM. Persistent faltering despite counselling needs a search for **HIV, TB, persistent diarrhoea/malabsorption, UTI** and a home visit or social assessment; treat persistent diarrhoea per UCG; refer to the paediatric clinic; if SAM criteria are met, start the SAM protocol.

Take-away: Measure, plot and act on the trend; most faltering is feeding-related, but HIV and TB must be excluded in Uganda.

Case 2: Not walking at 20 months

Kato, a 20-month-old boy, is brought because he 'does not walk yet'. He was born at 34 weeks at a HC IV, admitted for 2 weeks for 'yellowness and breathing problems'. He sits well, crawls, pulls to stand, says 'mama', 'baba', 'tata' and 'cup', and points to show things.

1. What is his corrected age, and is he delayed?

> Corrected age ≈ **20 – 1.5 = 18.5 months**. The limit age for walking is **18 months**, so he is at or just past the limit for walking; language (4 words, pointing) and social skills are about right. This is **isolated gross motor delay**: assess carefully.

2. What features in the history and examination would point to CP?

> Risk factors (prematurity, **neonatal jaundice**, respiratory problems); examination: increased tone in the legs, brisk reflexes, clonus, **scissoring**, toe-walking when supported, persistent primitive reflexes, asymmetry, involuntary movements (kernicterus). Also check hips, spine and feet.

3. What other causes of late walking will you consider?

> **Benign late walking** (bottom-shufflers, familial), **Duchenne muscular dystrophy** (boy: Gowers sign, calves, CK), neuromuscular disease (reflexes, fasciculation), **rickets** (bowing, wrist widening), developmental dysplasia of the hip, hypothyroidism, global delay (but his language is normal), lack of opportunity.

4. Examination: normal tone, reflexes and plantar responses; hips abduct fully; no bowing; he bears weight well and cruises. What do you advise?

> Likely **normal variation** (late walker) given normal neurology and other domains. Reassure, encourage practice (barefoot, push-along toy), measure growth, and **review in 2–3 months**; refer if not walking by about 2 years corrected, if new signs appear, or if any regression. Consider CK if a boy develops a Gowers sign.

5. What else must you check in this baby with a history of neonatal jaundice?

> **Hearing** (kernicterus causes auditory neuropathy), eye movements (upward gaze), teeth (enamel dysplasia), and subtle dystonia; his language is reassuring but formal hearing testing is wise.

Take-away: Interpret milestones with corrected age, by domain, and with a careful neurological examination; isolated late walking with normal neurology is often benign.

Case 3: The short 13-year-old

Atim, a 13-year-old girl from Lira, is referred because she is 'much shorter than her classmates'. Height 132 cm (HAZ –2.9), weight 28 kg. She has no breast development. Her father is 170 cm and her mother 160 cm.

1. Calculate her mid-parental height and interpret it.

> MPH = (170 + 160 – 13) ÷ 2 = **158.5 cm**; target range **150–167 cm**. Her current height is well below what her parents' heights predict, so **familial short stature is unlikely**.

2. What history will you take?

> Previous heights (school/clinic records) to estimate **velocity**; birthweight; chronic symptoms (diarrhoea, abdominal pain, cough, fevers); HIV status; sickle cell disease; diet and food security; headaches, visual problems, polyuria (pituitary); thyroid symptoms; school performance; family history of late puberty; recurrent ear infections or heart/kidney problems (Turner).

3. On examination she has a low posterior hairline, wide carrying angle, widely spaced nipples, Tanner B1 PH1, BP 128/80 in the right arm with weak femoral pulses. What is the likely diagnosis?

> **Turner syndrome (45,X)** with probable **coarctation of the aorta** (raised arm BP, weak femoral pulses). Turner syndrome causes short stature and **ovarian failure** (delayed puberty).

4. Which investigations will you request?

> **Karyotype** (Mulago/MakCHS genetics or private lab), **echocardiogram** (coarctation, bicuspid aortic valve), **renal ultrasound** (horseshoe kidney), TSH (autoimmune thyroiditis), LH/FSH (raised), bone age, hearing test; screen for coeliac disease and diabetes later.

5. What management does she need?

> Referral to **paediatric endocrinology** and **cardiology** (coarctation repair at the Uganda Heart Institute). **Growth hormone** improves final height (expensive, rarely available in Uganda); **oestrogen replacement** at about 11–12 years to induce puberty and protect bone; counselling about fertility (usually infertile; assisted reproduction with donor eggs), psychosocial support, school support.

> **Take-away:** Always compare a short child with the parents' heights and look at puberty; a short girl with delayed puberty needs a karyotype for Turner syndrome.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.